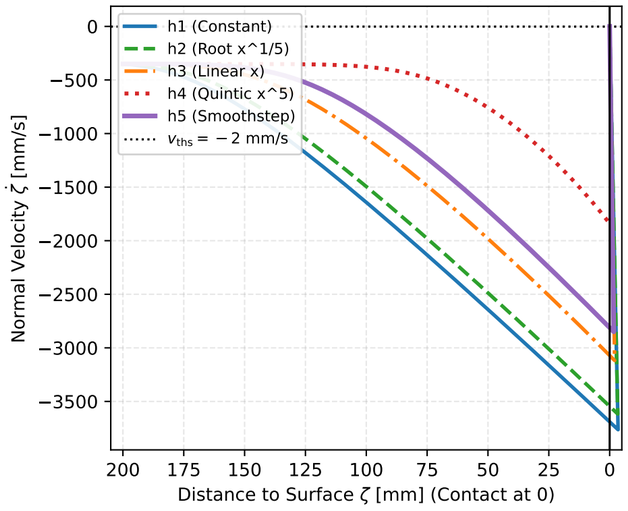
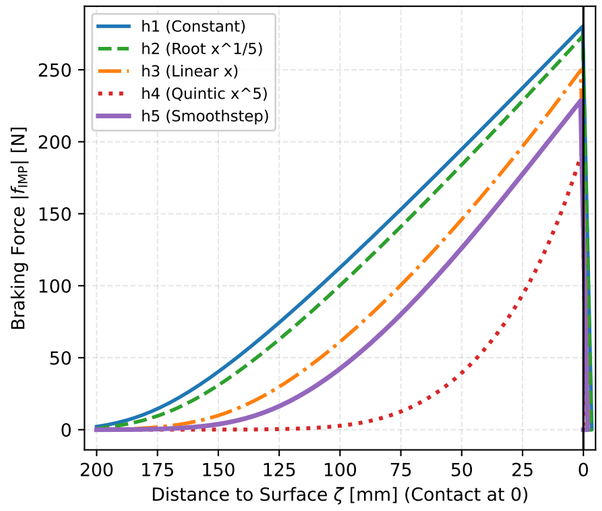
CBF Variable-Damping Cartesian Compliance Controller — Verification Suite Summary

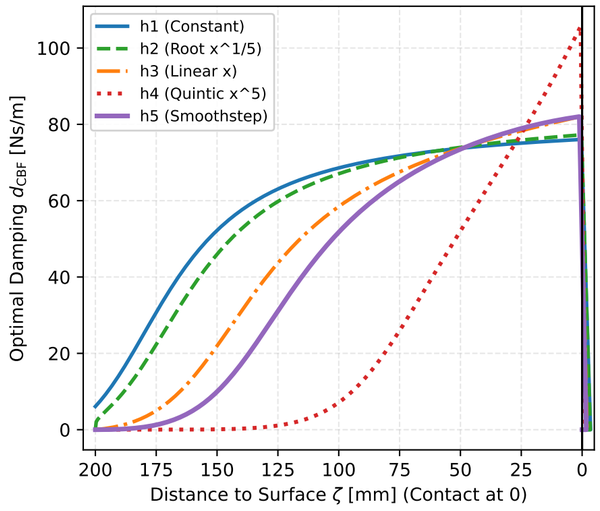
(1a) Phase Portrait: $\dot{\zeta}(\zeta)$ Envelope Shaping



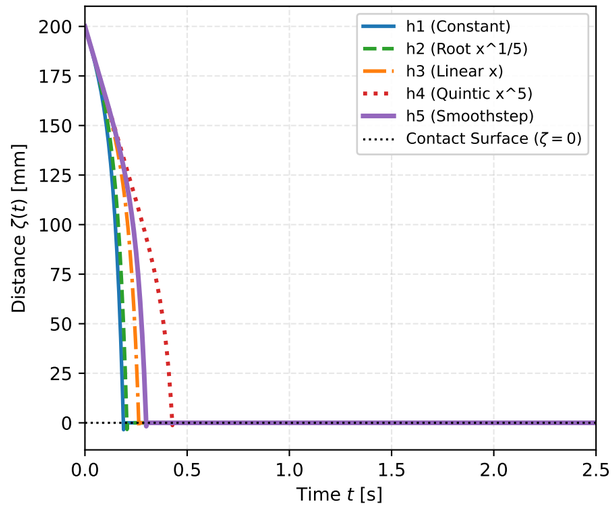
(1b) Spatial Force Distribution $|f_{IMP}(\zeta)| \propto h(u)$



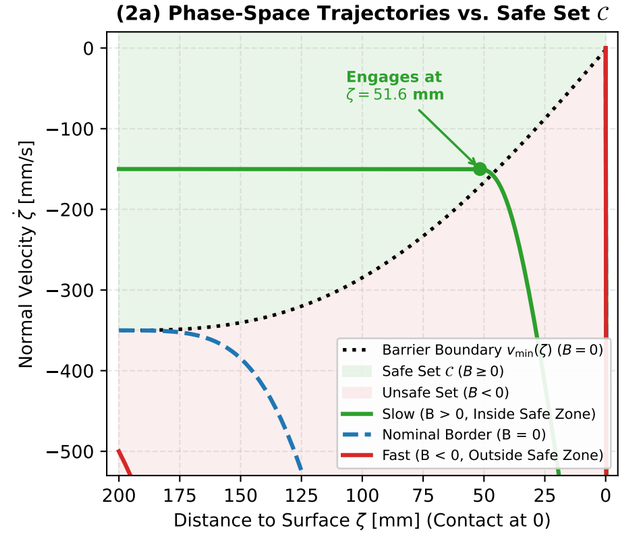
(1c) QP Damping Gain $d_{CBF}(\zeta)$



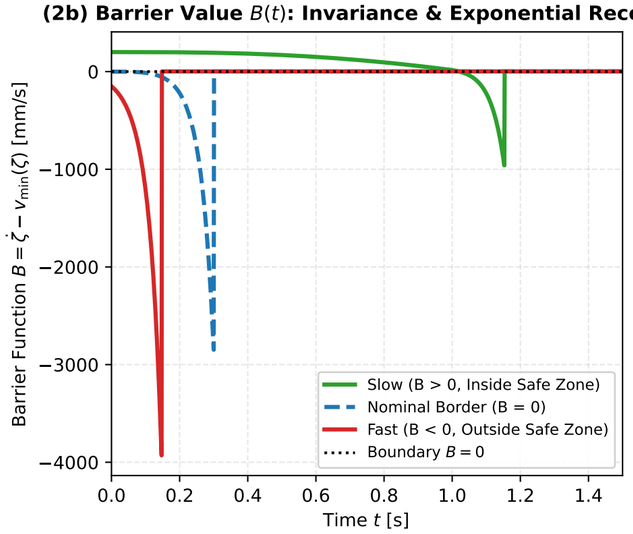
(1d) Approach Trajectory $\zeta(t)$ vs. Time



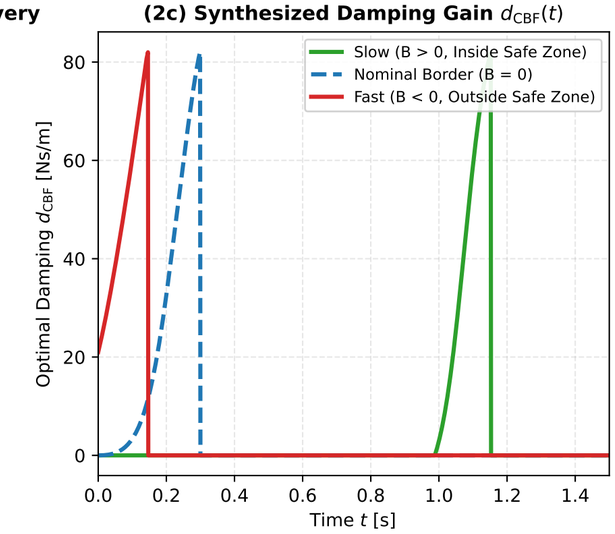
(2a) Phase-Space Trajectories vs. Safe Set $\mathcal{C}$



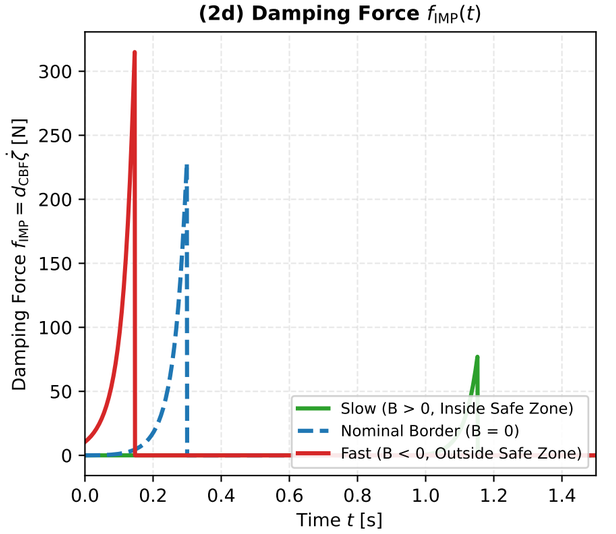
(2b) Barrier Value $B(t)$: Invariance & Exponential Recovery



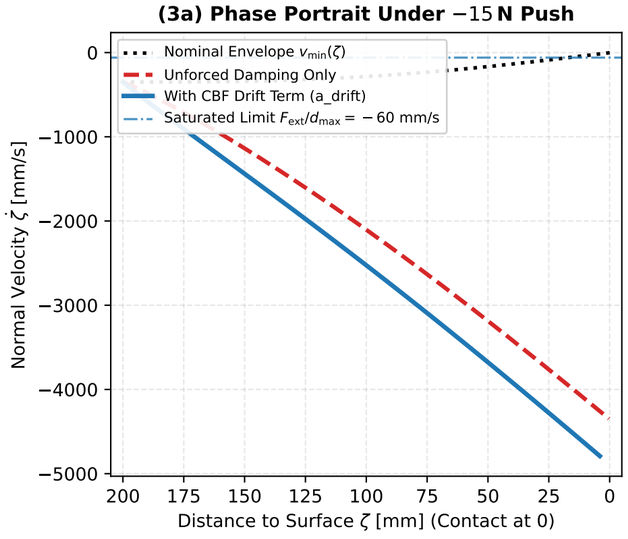
(2c) Synthesized Damping Gain $d_{CBF}(t)$



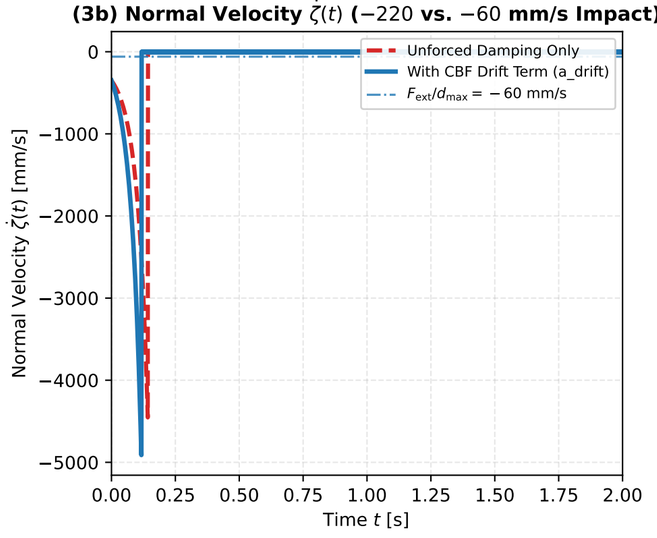
(2d) Damping Force $f_{IMP}(t)$



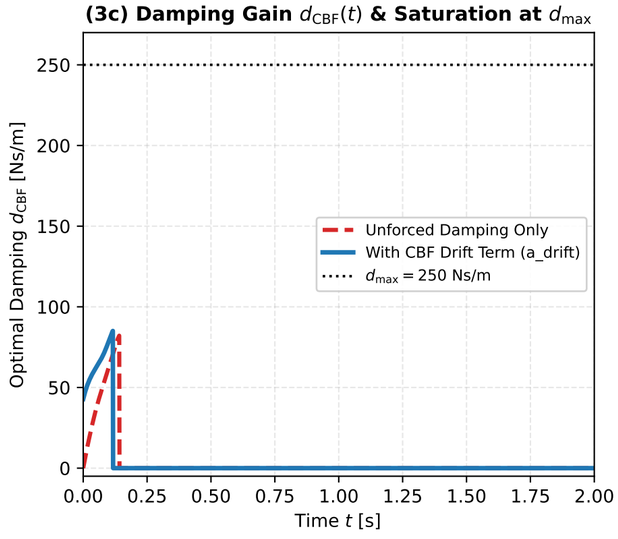
(3a) Phase Portrait Under -15 N Push



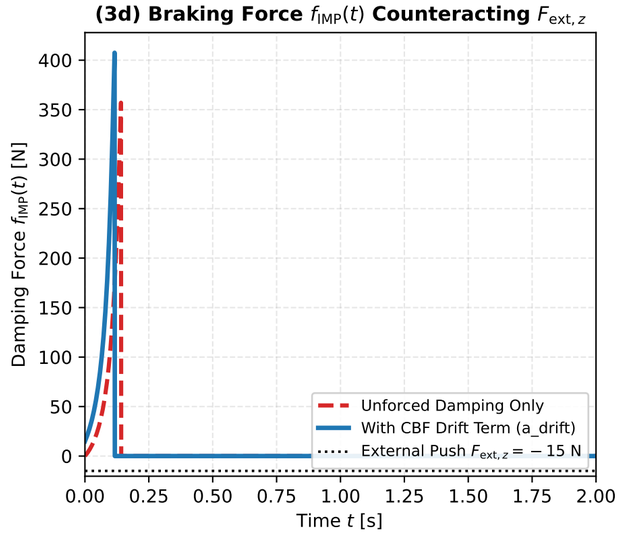
(3b) Normal Velocity $\dot{\zeta}(t)$ (-220 vs. -60 mm/s Impact)



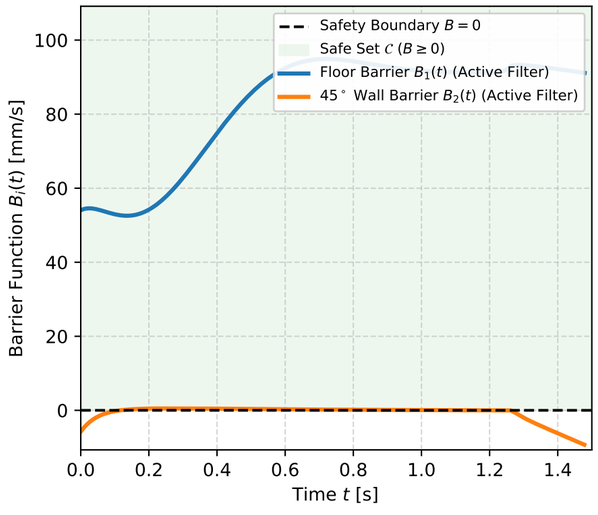
(3c) Damping Gain $d_{CBF}(t)$ & Saturation at d_{max}



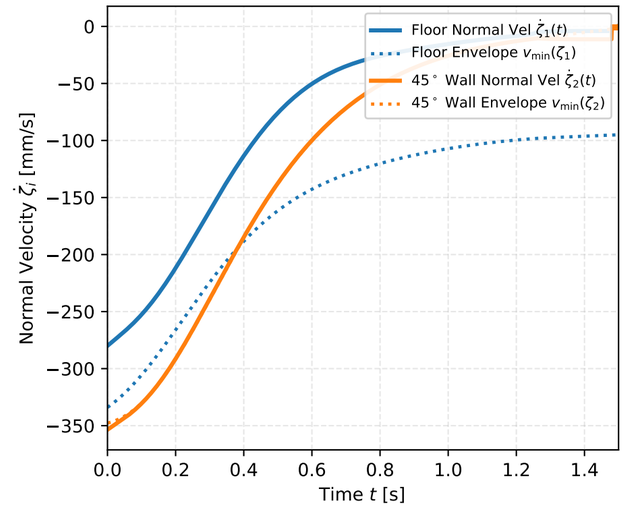
(3d) Braking Force $f_{IMP}(t)$ Counteracting $F_{ext,z}$



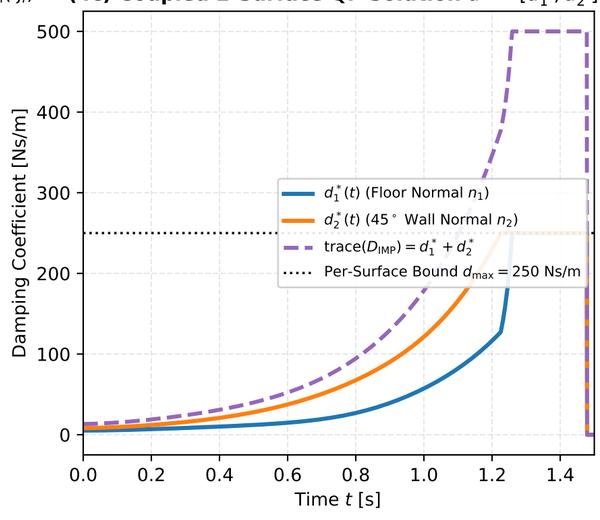
(4a) CBF Values $B_i(t)$ (Active Filter Intervals Only)



(4b) Multi-Surface Normal Velocities vs. Safe Bounds $v_{min}(\zeta_i)$



(4c) Coupled 2-Surface QP Solution $d^* = [d_1^*, d_2^*]^T$



(4d) End-Effector Surface Distances $\zeta_i(t)$

